

Implementing Clustering

Implement Clustering

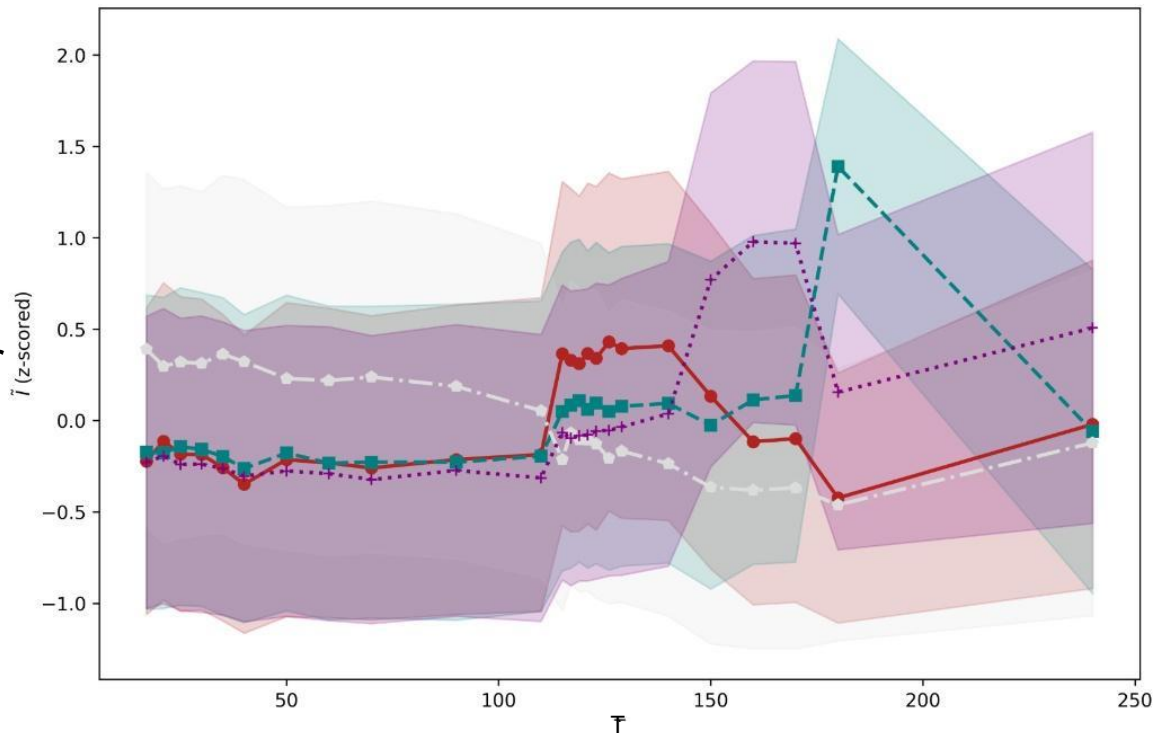
Algorithms

- K-Means
- DBSCAN
- GMM
- Agglomerative

Analyze how each algorithm suggests measurement strategies such as cluster centers, boundaries, or uncertain regions

K-Means

- Trajectories change at 110K-120K
- 110K-140K is where symmetry breaks
- Large shaded variance above 140K indicate either thermal noise or transitioning at different rates

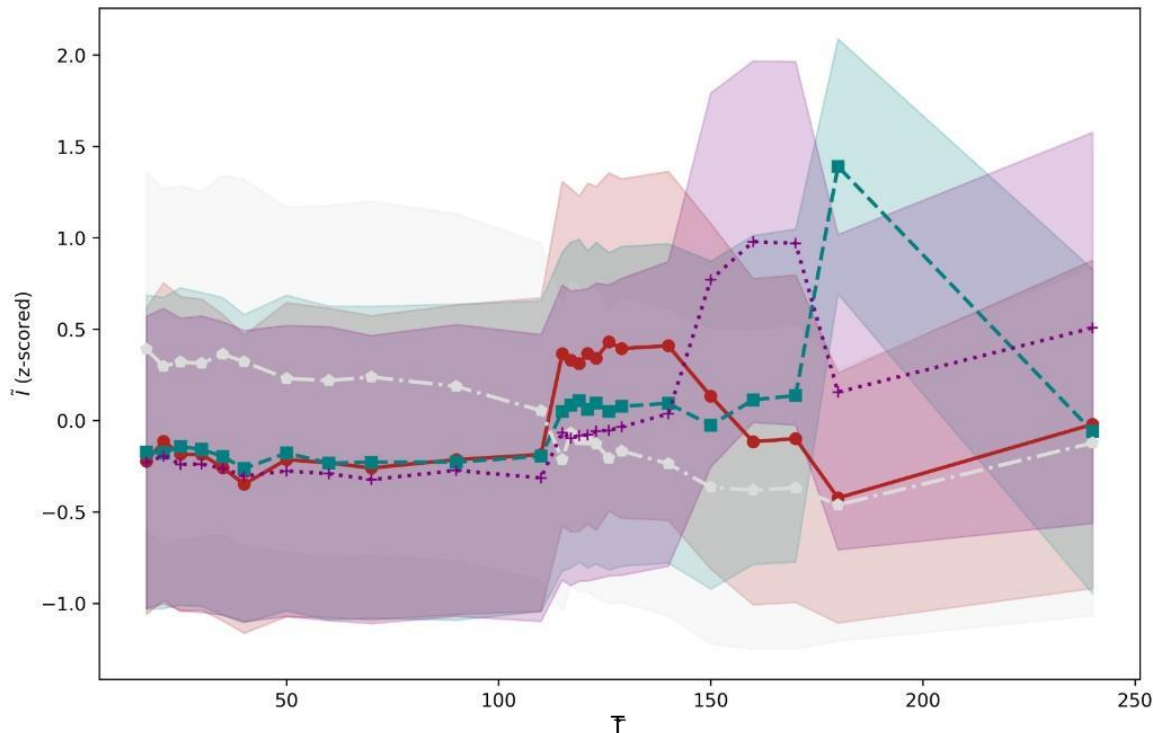


Measurement Strategies From K-Means

What it shows: Large shaded bands

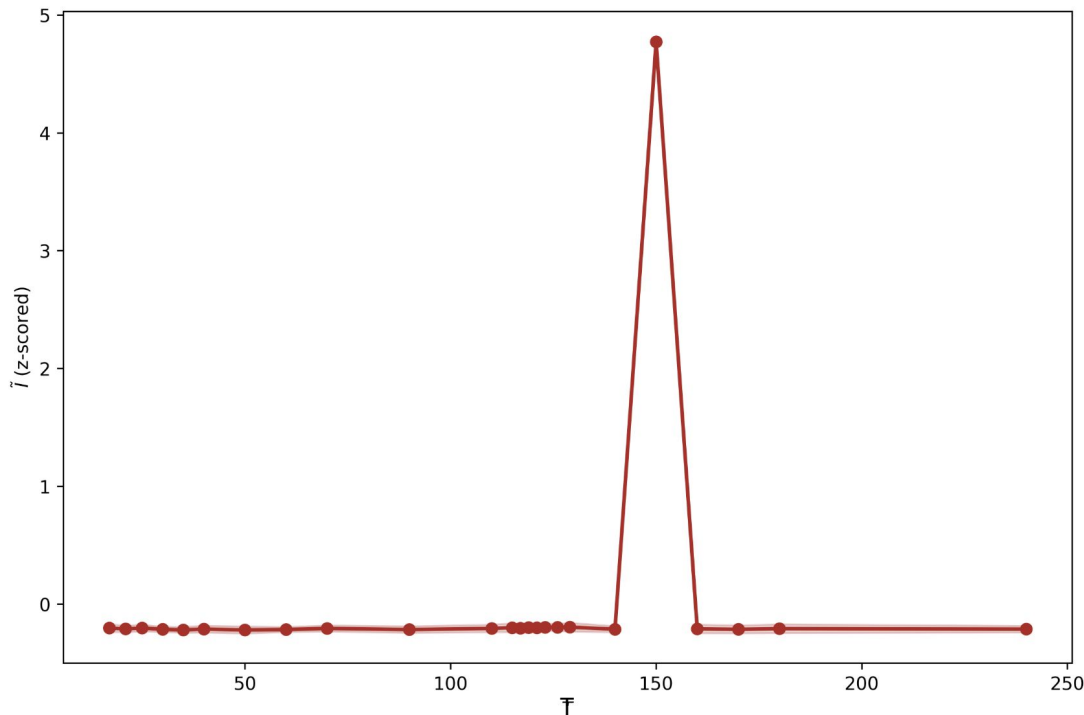
Strategy:

- Collect data for a longer time to get rid of possible noise



DBSCAN

- Trajectories fail to separate across all temperatures
- Algorithm groups data into single flat data due to lack of empty density gaps
- Spike at 150K is flagged as a outlier

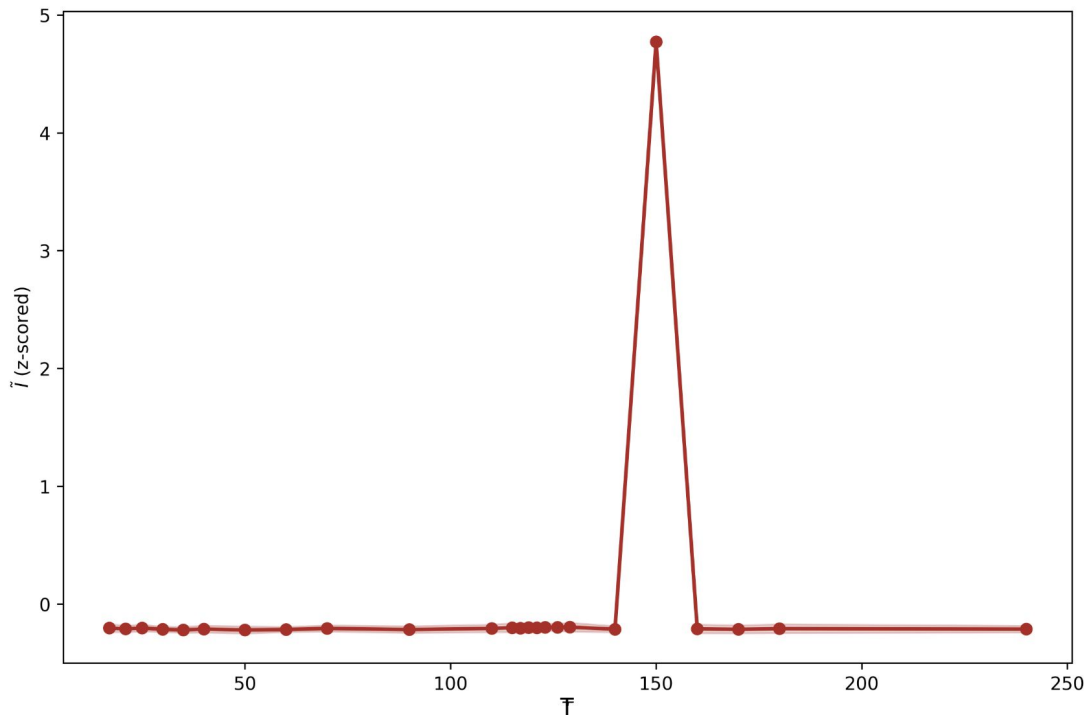


Measurement Strategies From DBSCAN

What it shows: All data clustered into one group and a spike at 150K categorized as noise

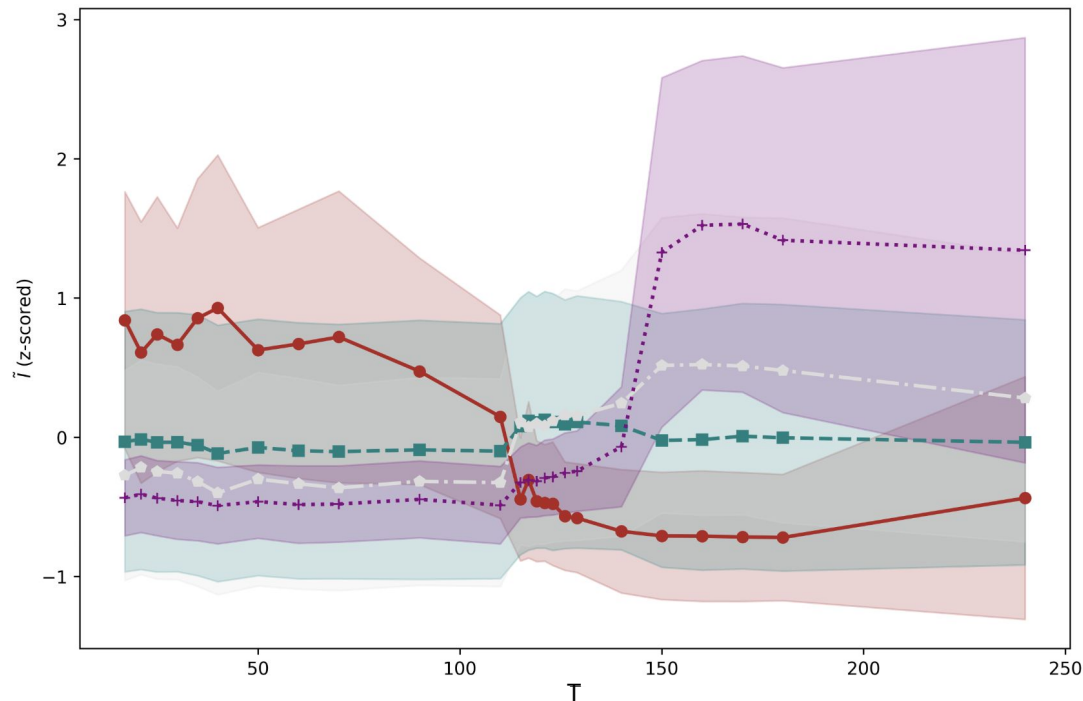
Strategy:

- Double check the noise point
- Use a different clustering algorithm



GMM

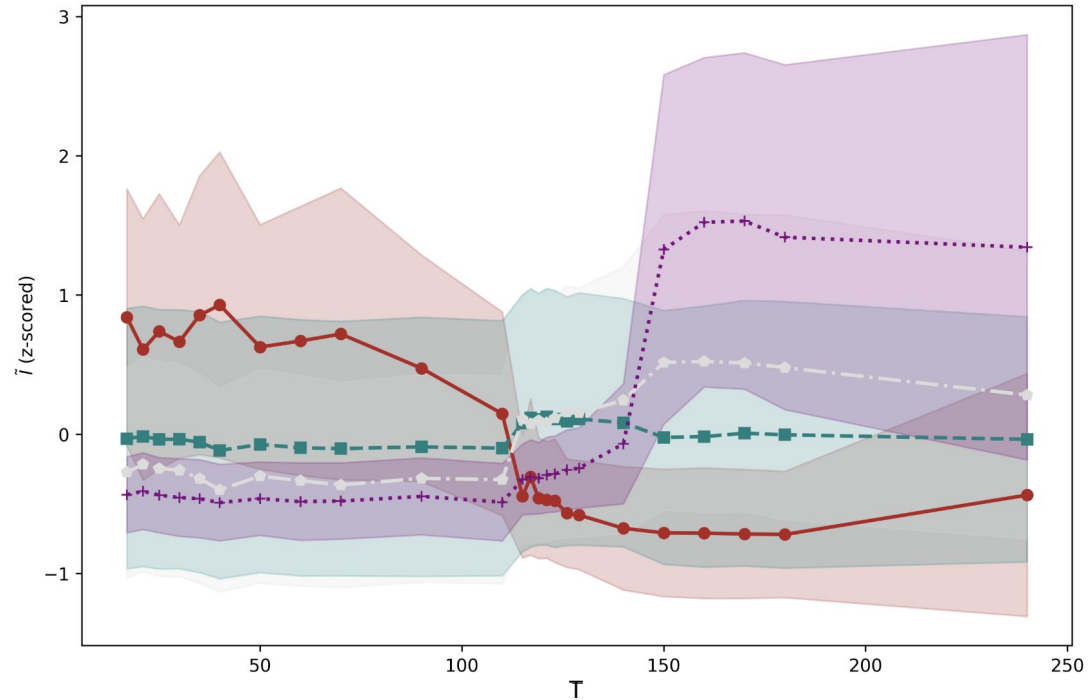
- Trajectories change at 110K-120K
- Physical crossover occurs at 140K
 - Red and purple phases swap
- Large shaded bands above 140K indicates overlapping crystal phases and noise



Measurement Strategies From GMM

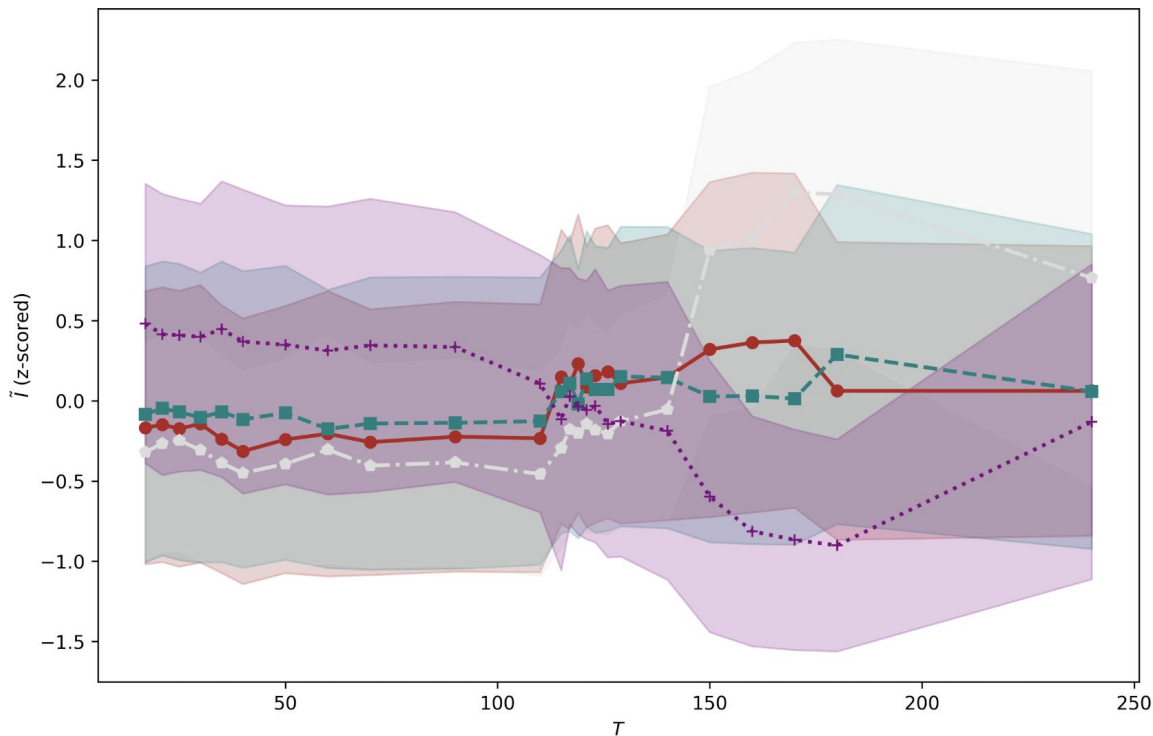
What it shows: Above 140K, the large shaded bands show high uncertainty since the crystal is split into different domains

Strategy: Use a smaller X-ray spot size about 140K to zoom in on one domain at a time



Agglomerative

- Trajectories separate at around 110K-120K
- At 140K rearrangement occurs
 - Purple phase drops while red and white phases climb
- Large shaded variance shows AGNES separates phases even at low temperatures



Measurement Strategies From Agglomerative

What it shows: Below 110K, phases are already split

Strategy: Spend more measurement time below 110K to find out why the purple path starts out separated

